Curriculum Vitae

Mohamed Salaheldeen

Institut de Mathématiques de Bourgogne (IMB)

mohamed-salaheldeen.mohamed@ube.fr

Education

Institut de Mathématiques de Bourgogne (IMB) — Dijon, France
M2 Research Master in Mathematical Physics, Grade: 16.8/20 (Rank 1) — Sep 2024 – May 2025
Master Thesis: First Contact with Factorization Homology
Advisor: Prof. Lukas Woike

The Abdus Salam International Centre for Theoretical Physics (ICTP) — Trieste, Italy
Postgraduate Diploma in High Energy Physics — Sep 2022 – Aug 2023
Thesis: 2D Topological Field Theories
Advisor: Prof. Pavel Putrov

University of Science and Technology at Zewail City — Giza, Egypt
Bachelor of Science, Physics, GPA: 3.87/4.0 (Highest Honors) — Sep 2018 – Jun 2022
Graduation Thesis: Spherically symmetric solutions in modified Teleparallel gravity
Advisor: Prof. Alexey Golovnev

Research Experience

Institut de Mathématiques de Bourgogne (IMB) — Dijon, France
Doctoral Researcher — Oct 2025 - Present
Advisor: Prof. Peter Schauenburg
Topic: Orthogonality for group-theoretical fusion categories.

Bogoliubov Laboratory of Theoretical Physics (JINR) — Dubna, Russia
Research Trainee (Numerical Methods) — Sep 2021 – Nov 2021
Advisor: Prof. Yakov Shnir
Project: Numerical construction of topological solitons in scalar field theories.

Center of Fundamental Physics — Giza, Egypt
Research Intern (General Relativity) — Aug 2021 – Sep 2021
Advisor: Prof. Shaaban Khalil
Project: Symmetries in Einstein Field Equations for the derivation of Schwarzschild and Kerr metrics.

Seminar and Research Talks

Seminar Organization

2026 – Present: Organizer, Working Group on Stable Homotopy Theory and Higher Algebra, IMB

Invited & Contributed Talks

12 May 2026: Modular Categories and Construction of WRT Invariants (Notes), Topology Between the Waves VI

7 May 2026: Positive geometry, Working Group on Geometry & Physics, IMB

10 Apr 2026: The WRT 3d-TQFT from modular fusion categories, Working group on TQFTs (non-)semisimples, IMB

10 Mar 2026: Introduction to Quantum Groups, PhD Seminar, IMB

Selected Schools & Conferences

Aug 2026 – Sep 2026: Summer School CRC 1624: Higher structures, moduli spaces and integrability, Universität Hamburg, Germany

Jun 2026: GCS '26 School (Global Categorical Symmetries), Institut Henri Poincaré, Paris, France

Jun 2026: Masterclass on Seiberg-Witten Theory and 4-Manifolds, Univ. of Copenhagen, Denmark

May 2026: Spring School "Khovanov and Floer Homotopy", Nantes Université, France

May 2026: Topology Between the Waves VI, Fréjus, France

Mar 2026: UNIVERSE+ School: Exploring Positive Geometry, Les Houches School of Physics, France

Dec 2025: Masterclass on Shifted Symplectic Structures, Centre Henri Lebesgue, Angers, France

Dec 2024: Masterclass on Graph Complexes, Centre Henri Lebesgue, Angers, France

Teaching Experience

Department of Applied Mathematics, Zewail City — Giza, Egypt

Teaching Assistant — Oct 2023 – Jun 2024

Courses: Ordinary Differential Equations, Partial Differential Equations, Complex Analysis, Discrete Mathematics, Numerical Analysis.

Awards & Honors

2018 – 2022: Provost's Honor Roll, Zewail City (8 consecutive semesters, GPA > 3.7)

Dec 2020: Finalist, XI International Natural Sciences Tournament, Moscow, Russia

Sep 2016: National Finalist, International Physics Olympiad (IPhO), Nile University

2016: Micro Research Grant (MRG), Center of Special Studies and Programs, Bibliotheca Alexandrina

Skills

Languages: Arabic (Native), English (C1), French (B1)

Computational: Python, C++, Mathematica, ROOT Framework, LaTeX